VAYU | YOUR PERSONAL CROP GUIDE

Pomegranate

How to grow it on your land

Prepared for: General guide (no location chosen)
Prepared on 2026-11-11

Suits hot, dry, semi-arid land with drip irrigation, and it earns well. But bacterial blight has made many growers uproot orchards, so disease-free planting material and strict hygiene decide success.

DIFFICULTY	MONEY NEEDED	RISK	FIRST RETURN
4 of 5	**Medium**	**High**	**fruiting from about the 4th year**

How to read this guide: [S1] marks a line backed by a source listed on the last page. (general knowledge) marks a line that has not been verified against a source. Lines starting **For your land** use the numbers for your location.

In this guide

- Your priorities: what to do first, in order
- Your planting plan: how much to buy and prepare for your plot
- How to grow it: planting, water, nutrition, pests and diseases, monitoring, harvest, risks
- If your conditions are different, and the sources behind every line

Your land at a glance

Your priorities

Ordered by what matters most on your land.

1. **Get your soil tested**
Satellite and model data estimate soil pH and texture only roughly, and cannot see nutrients, salinity or soil health. Get a proper soil test (pH, organic carbon, nutrients, salinity) from your KVK or the state soil-testing laboratory; the Soil Health Card scheme provides this. (general knowledge)

2. **Talk to your KVK first**
Take this guide to your nearest Krishi Vigyan Kendra and ask for their local advice and trusted planting material. (general knowledge)

3. **Start small**
Plant or build a small trial first. Learn the crop on your own land, then expand from what actually worked. (general knowledge)

4. **Line up a buyer before you scale**
Know who will buy, in what form and grade, and how it will reach them. (general knowledge)

Your planting plan for 1 acre

What	How much	Basis
Trees at 3 m x 3 m	**about 449**	your plot size x cited amounts [S1]

Sums of the cited per-plant and per-hectare amounts for your plot size. Real needs vary with soil and variety; confirm with your KVK.

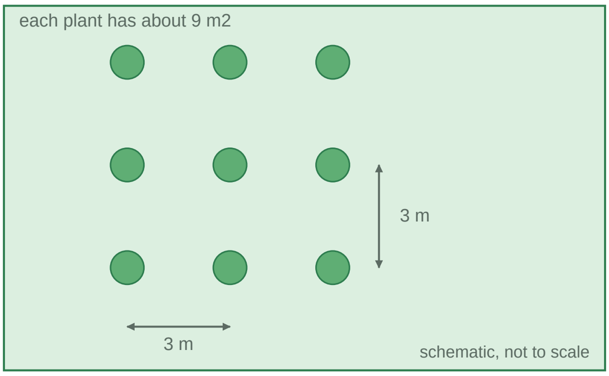
Planting material

- Use certified, disease-free, true-to-type plants free of nematodes, wilt and bacterial blight; tissue-cultured plants from approved sources are one option. ICAR-NRC on Pomegranate (Solapur) is the reference institute. [S1, S8]
- Infected planting material is how bacterial blight reaches new areas, so never buy cheap unverified saplings. [S8]
- Jalore Seedless and G-137 (bred for blight tolerance) are among varieties named, besides Bhagwa; ask your KVK what works locally. [S7, S9]

Land preparation and planting

- Deep alluvial or sandy to sandy-loam soil is best. Plant at about 3 m x 3 m (about 1,111 plants per hectare for Bhagwa); some advice uses wider 4.5 m x 4.5 m to improve airflow. [S1, S6]
- Plant in the monsoon (June-August) or February-March. [S9]

- Mulch the root zone 15-20 cm thick with weeds and dry grass. [S1]

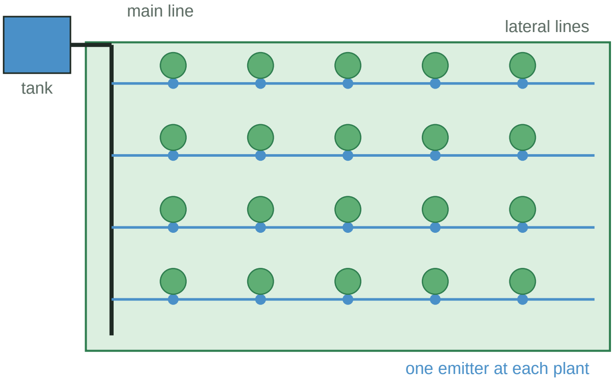

each plant has about 9 m2
3 m
3 m
schematic, not to scale

Plant spacing, top view

Irrigation

- Drip irrigation is standard. A drip trial at ICAR-NRCP gave tree water use of about 5.7-13 litres per day in year 2 rising to 10-23 litres in year 5 for Mrig bahar. [S10]

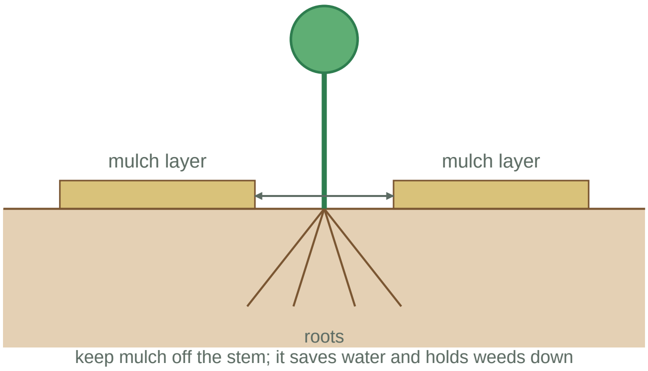
- Crop regulation (bahar) uses planned water withholding, for example April-May for the Mrig bahar crop; learn the schedule from a trained source before trying it. [S11]


main line
lateral lines
tank
one emitter at each plant

Drip irrigation layout (schematic)

Nutrition (organic)

- Farmyard manure or compost around the drip line each year, with mulch; this guide does not give chemical fertiliser doses. Use a soil test and your KVK's schedule. (general knowledge)

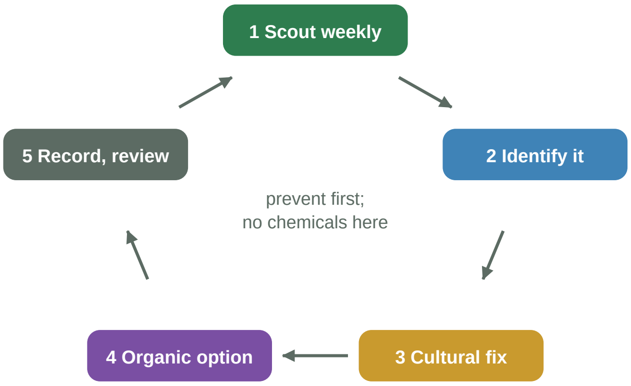

mulch layer
mulch layer
roots
keep mulch off the stem; it saves water and holds weeds down

Mulch around a plant

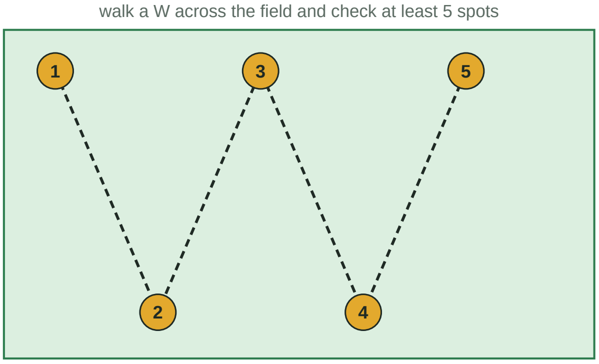
Pests and diseases (organic and cultural only)

- Bacterial blight is the most damaging disease. Prevent with disease-free plants, wide spacing and airflow, and removing infected fruit and twigs and burying them in a covered pit. [S6, S12]
- Biological options studied include Pseudomonas fluorescens and Bacillus subtilis; ask your KVK for a registered biocontrol product and method. [S6]
- Use drip, not sprinklers, so the canopy stays dry; disinfect pruning tools between trees. (general knowledge)
- Institute advisories do list chemical sprays; this guide does not reproduce them. Get the current local schedule from ICAR-NRCP or your KVK. [S12]

- For anything beyond these organic and cultural steps, ask the nearest Krishi Vigyan Kendra (KVK) or state agricultural university for their current recommendation. (general knowledge)



Pest and disease loop: prevent first, then look, fix, review



How to walk the field when scouting

Monitoring

- Walk the orchard weekly: look for dark leaf spots, oily fruit spots, cracking and wilting branches, and remove affected material immediately. (general knowledge)
- Keep a log of flowering, fruit set, irrigation and disease sightings per tree block. (general knowledge)

Harvest, yield and post-harvest

- Fruiting from about the 4th year; harvest timing depends on the bahar (for Mrig bahar, December-January at Solapur). [S13, S11]
- Improved management including blight control raised net returns about 25 percent over farmer practice in an Andhra Pradesh trial of 50 farmers. [S13]

Risks and costs

- A reported 30 percent or more of farmers uprooted trees over four years because of blight, wilt and similar diseases. [S6]
- Water supply and a reliable market and grading are the other big risks; plan both before planting. (general knowledge)

If your conditions are different

What to do if your land, or the season, turns out different from the numbers above.

- **Heavy soil** (fixable): Deep alluvial or sandy to sandy-loam soil is best; heavy clay holds water and encourages disease. Plant raised, use drip irrigation, and keep the basin drained. [S1]
- **Heat** (fixable): Pomegranate endures up to about 48 C with hot dry winds; fruit colour and quality are best in a hot dry fruiting season. Heat is rarely the limit; water is. [S1]
- **Low rainfall** (fixable): Water is the limiting factor in arid regions; use drip irrigation, mulch 15-20 cm thick, and plan the water budget before planting. [S1, S4, S5]
- **Too much rain** (fixable): Humid, rainy, cloudy fruiting seasons raise disease (bacterial blight, fruit spots) and cracking. Prefer a hot dry fruiting season; choose your flowering season (bahar) to avoid heavy rain on fruit. [S6,

S1]

- **Cold weather** (fixable): It is deciduous and tolerates cold winters when irrigated, but young trees can be damaged by hard frost; protect them in the first winters. (general knowledge)
- **Soil pH** (fixable): Ideal pH is about 6.5-7.0, but good orchards in Solapur grow in alkaline soils (pH around 7.8-8.8). Add compost; avoid saline patches. [S1, S7]

About chemicals

This guide gives organic and cultural methods only. It does not name or dose pesticides, fungicides or chemical fertilisers; for those, follow your nearest Krishi Vigyan Kendra (KVK) or state agricultural university and the label of any registered product.

Sources

[S1] https://hpshiva.hp.gov.in/cms/media/ggojam4f/pomegranate.pdf
[S4] https://epubs.icar.org.in/index.php/IJAgS/article/download/65400/27693/164294
[S5] https://www.phytojournal.com/archives/2020/vol9issue1/PartH/8-3-661-262.pdf
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[S12] https://nrcpomegranate.icar.gov.in/files/Advisory/71.pdf
[S13] https://www.phytojournal.com/archives/2020/vol9issue1/PartH/8-3-661-262.pdf

Advisory aid from published sources and general agronomic knowledge, not a guarantee of yield or income and not a substitute for local extension advice. Lines marked 'general knowledge' are not verified against a source.